Name

Mathematics Stage 5 Paths

Simple Cubic Equations

Book 1

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Learning Intention ..... **Error! Bookmark not defined.**

Syllabus Content

MA5-EQU-P-01 solves monic quadratic equations, linear inequalities and cubic equations of the form $ax^3 = k$

Solve cubic equations

- Determine that for any value of k there is a unique value of x that solves a cubic equation of the form $ax^3 = k$ where $a \neq 0$
- Solve cubic equations of the form $ax^3 = k$, leaving answers in exact form and as decimal approximations

Simple Cubic Equations

Simple Cubic Equations

- **Cubic equations** have unknowns to the power of 3: x^3 "x cubed"
- **Cube rooting** $\sqrt[3]{\ }$ is the inverse operation of cubing
- A cubic equation in the form $ax^3 = k$ always has **one** real solution.
- Both positive and negative numbers can be cube rooted.

- To solve cubic equations:
  1. Isolate x^3
  2. Cube root both sides.

Example Solve cubic equations in the form ax^3=k

$x^3 = 27$

2. Cube root $\sqrt[3]{\ } = \sqrt[3]{\ }$

$x = \sqrt[3]{27}$

$x = 3$

$-2x^3 = 16$

1. Isolate x^3 $\div -2 \quad \div -2$

$x^3 = -8$

2. Cube root $\sqrt[3]{\ } = \sqrt[3]{\ }$

$x = \sqrt[3]{-8}$

$x = -2$

Guided Practice

Solve these equations.

a $x^3 = 64$

$x = 4$

b $x^3 = 125$

$x = 5$

c $x^3 = -27$

$x = -3$

d $x^3 + 2 = -6$

$x = -2$

e $2x^3 = 2000$

$x = 10$

f $\frac{x^3}{2} = 108$

$x = 6$

Key Ideas

1. The inverse operation of cubing is …… ……

1 Solve the following. Round your answer to 2 d.p, if appropriate.

a	$x^3 = 8$	**b**	$x^3 = -64$	**c**	$x^3 = 216$
	$x = 2$		$x = -4$		$x = 6$
d	$x^3 = 27$	**e**	$x^3 = -1000$	**f**	$x^3 = -125$
	$x = 3$		$x = -10$		$x = -5$
g	$27x^3 = 8$	**h**	$125x^3 = -27$	**i**	$27x^3 = 125$
	$x = \frac{2}{3}$		$x = -\frac{3}{5}$		$x = \frac{5}{3}$
j	$8x^3 = 8$	**k**	$7x^3 = 8$	**l**	$-8x^3 = 11$
	$x = 1$		$x = \sqrt[3]{\frac{8}{7}} \approx 1.05$		$x = \sqrt[3]{-\frac{11}{8}} \approx -1.11$
m	$n^3 = -184.7$	**n**	$b^3 = \frac{125}{8}$	**o**	$r^3 = -0.343$
	$x = \sqrt[3]{-184.7} \approx -5.69$		$b = \frac{5}{2}$		$r = -0.7$
p	$5x^3 = 135\,000$	**q**	$-8n^3 = 569$	**r**	$\frac{k^3}{4} = 65$
	$x = 30$		$n = \sqrt[3]{-\frac{569}{8}} \approx -4.14$		$k = \sqrt[3]{260} \approx 6.38$
s	$\frac{2t^3}{5} = -3200$	**t**	$3r^3 - 41\,472 = 0$	**u**	$32 - \frac{x^3}{9} = -48$
	$t = -20$		$r = 24$		$x = \sqrt[3]{720} \approx 8.96$
v	$7(y^3 - 2) = 2183$	**w**	$6y^3 = 4y^3 + 45$		
	$y = \sqrt[3]{\frac{2197}{7}} \approx 6.80$		$y = \sqrt[3]{\frac{45}{2}} \approx 2.82$		

2

a Write the volume of a cube with side x cm:

$V = x^3$

b The volume of three cube shaped containers is 18 m^3.
Calculate the side length of one of the cubes in exact form.

$18 = 3x^3$

$x = \sqrt[3]{6}$ cm

3 A rectangular prism has dimensions in ratio 1:2:3.
Calculate the side lengths if the volume is 4374 cm^3.

$x \times 2x \times 3x = 4374$

$x = 9$

$\therefore 9 : 18 : 27$ cm